

BUSINESS PROPOSAL

Project Zync

The Sovereign AI for Rare Earth Engineering

UMHackathon 2026 Domain:

AI for Economic Empowerment & Decision Intelligence

Team: Gunners

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1. Executive Summary:

*Zync is Malaysia's first **Sovereign AI Decision Intelligence Tool** designed to unlock the RM 1 trillion potential of the nation's 16.1 million metric tonnes of Non-Radioactive Rare Earth Element (NR-REE) reserves. Despite the 2024 raw REE export ban under the 13th Malaysia Plan (RMK13), Malaysia faces a **30-year "Intelligence Gap"** in metallurgical engineering, forcing dependency on foreign technological monopolies.*

*Zync bridges this gap by providing site engineers with a seven-agent AI pipeline—powered by Z.AI's GLM-5.1—that digitizes handwritten field logs, optimizes chemical leaching flowsheets, and ensures real-time regulatory compliance. By compressing diagnostic timelines from 3 weeks to under 30 minutes and reducing technical consultation costs from **RM 80,000** to pennies per session, Zync empowers Malaysia to transition from a raw material extractor to a high-value processing hub.*

*Zync is not just a software tool; it is a **Strategic Reserve of Intelligence**. By deploying it now, Malaysia ensures that when the 13th Malaysia Plan concludes in 2030, the nation owns both the minerals and the methodology to refine them.*

2. Product Overview / Solution

Key Features:

- **Evidence Diagnosis Module:** Uses multimodal vision to digitize handwritten field logs and identify root causes for yield drops in seconds.
- **Process Optimization Module:** A "Chemical Architect" that runs up to 30 iteration loops to design optimal leaching flowsheets based on first principles.
- **Zone Prioritisation Module:** A strategic tool for planning analysts to rank candidate extraction sites based on economic value vs. ESG risk.
- **Regulatory Block Card:** An automated safety trigger that halts operations if designs risk a thorium violation

Technology Used:

- **Model:** Z.AI GLM-5.1 for scientific reasoning and multi-agent orchestration.
- **Data Layer:** Neo4j GraphRAG for retrieving historical Malaysian geological precedents via traversal.
- **Backend:** Python-based multi-agent pipeline with StreamingResponse for real-time reasoning traces.
- **Infrastructure:** Sovereign Malaysian Cloud to ensure data residency and national security.

3. Market Research & Market Dynamics

3.1 Problem Landscape

*The "Intelligence Gap" is the primary barrier to Malaysia's RM 1 trillion opportunity. Traditional flowsheet development relies on physical bench testing costing roughly **RM 80,000** per engagement and takes weeks of iteration. Furthermore, 91% of global refining capacity is currently controlled by China, leaving Malaysia vulnerable to foreign IP monopolies.*

3.2 Market Niche

- **Primary:** Site engineers at the 20+ active Malaysian IAC-REE facilities (e.g., Gerik, Perak).
- **Secondary:** State Development Units (e.g., MBI Perak, KSSB Selangor) requiring regulatory oversight tools.
- **Institutional:** NRECC and JMG policy advisors monitoring RMK13 Strategy A1.6 alignment.

3.3 Market Sizing

Market Tier	Definition	Estimated Value (2026)
TAM	Asset Value of Malaysia's 16.1M Tonne Reserves	RM 1,000 Billion
SAM	Cumulative Technical Consultant Spend (20 Active Sites)	RM 160 Million / Year
SOM	Initial Capture of State-Level Pilot Projects (Perak/Kedah)	RM 12.4 Million

3.4 Market Dynamics & Placement:

- **Regulatory Forcing Function:** The raw REE export ban creates a captive market for domestic refining tools.
- **Geopolitical "De-sinicization":** The U.S.-Malaysia MOU on Critical Minerals (Oct 2025) drives demand for non-Chinese metallurgical intelligence.
- **Digital MADANI Pillar:** RMK13 explicitly mandates the use of AI to increase economic complexity in the mineral sector

4. Value Proposition & Competitive Advantage

Zync redefines the economics of metallurgy by replacing **RM 80,000** consultant engagements with **RM 0.50** AI sessions.

- **Scientific Transparency:** Unlike "black-box" foreign software, Zync provides a visible reasoning trace, acting as a training tool for local engineers.
- **Regulatory "Hard Gates":** Zync is the only tool with a real-time Compliance Agent (Agent 4) that automatically blocks violations of AELB 1986 and DOE EQA 1974.
- **Sovereign Data Fortress:** All geological reasoning is grounded in local Malaysian data, ensuring that the "Sovereign Brain" of the refinery remains under national control.

5. Competitor Analysis (Porter's Five Forces)

Force	Intensity	Assessment
Competitive Rivalry	High	Driven by global giants like Lynas expanding into HREE and Chinese SOEs protecting their 91% refining monopoly.
Threat of New Entrants	Low	Requires high technical depth in both deep metallurgical knowledge and high-tier AI orchestration.
Bargaining Power of Buyers	Medium	Malaysian site owners have the resource (RM 1T) but lack the "brain" to process it, giving Zync high leverage.
Bargaining Power of Suppliers	Low	Cloud and LLM costs are commoditizing; the true "supplier" power lies in the unique Malaysian geological data Zync holds.
Threat of Substitutes	Medium	The "substitute" is the RM 80,000 human consultant, whom Zync displaces through 99% cost reduction.

Competitor Comparison Table:

Competitor	Strength	Weakness	Zync's Advantage
METSIM	Industrial standard for mass and energy balance.	Requires manual expert-defined parameters; no generative reasoning.	Generative Intelligence: Provides "starting recipes" via SciGLM reasoning in seconds.
Chinese SOEs	Decades of operational data from similar Jiangxi deposits.	Does not fit Malaysian IAC-REE Profiles. No knowledge transfer; creates a permanent state of technological dependency.	Sovereign & Explainable: Localized AI that shares its "reasoning trace" with Malaysian engineers.
Lynas	Fully integrated "Mine-to-Magnet" ecosystem in Pahang.	Proprietary expertise is focused solely on their own closed-loop supply chain.	Democratized Access: Accessible to independent and state-level miners at a fraction of the cost.

6. Business Model Canvas

6.1. Customer Segmentation

1. **Site Operators:** Focus on yield recovery and cost reduction.
2. **State Regulators:** Focus on ESG compliance and river proximity safety.
3. **Institutional Investors:** Focus on "Zone Scoring" for capital mobilization.

6.2. Revenue Stream

Stream	Price Point	Target	Purpose
Pilot License	RM 5,000 (One-time)	Junior Miners / Exploration Teams	Low-Friction Trial: Proves the AI's accuracy on local IAC-REE profiles before a major commitment.
Annual Subscription	RM 36,000 / year	Active Production Sites (Private Sector)	SaaS Baseline: Provides "Co-Pilot" intelligence for daily yield management and AELB/DOE compliance.
Sovereign Enterprise	RM 500,000 / year	State MBIs & Federal Regulators	Governance: A "Command Center" for state-level oversight across all private miners in their territory.
Yield Share	2% of Proven Yield Increase	All Active Producers	The Performance Tax: Captures a slice of the RM 1 trillion value created by the AI's "Smart" reasoning.

6.3. Cost Structure

Cost Drivers	Cost Type
R&D for the multi-agent pipeline, Neo4j graph maintenance.	Fixed Cost
GLM API tokens (~RM 0.50 per full pipeline run).	Order-based / Variable Cost
Cloud hosting on sovereign Malaysian servers.	Overhead Cost

6.4. Key Partners

Internal	External
Z.AI (Model provider)	MBI Perak (Data/Pilot partner)
UM Department of Geology (R&D)	JMG (Regulatory alignment)

6.5 Risk Assessment

Risk Type	Impact	Likelihood	Mitigation
Thorium Concentration Violation	High	Low	Agent 4 Safety Gate: Automatically halts lixiviant design iterations if the predicted Thorium co-extraction exceeds 0.8 Bq/g (20% safety margin).
Hydraulic Solution Leakage	High	Low	Agent 6 Hard-Logic: Uses geological PDFs to calculate "weathered crust thickness" and enforces a 500m "Hard Buffer" from all permanent river systems.
"Intelligence Sabotage"	Medium	Medium	Sovereign Brain Protocol: Every recommendation includes a full "Reasoning Trace" in English/Bahasa Malaysia, ensuring the IP remains with Malaysian engineers.
Subsurface Strata Misalignment	High	Medium	Diagnostic Agent Vision: Interprets subsurface medium wave velocity and operator notes to adjust injection depth in real-time.

7. Implementation & Financials

7.1 Go-to-Market Strategy

- **Phase 1: The Perak Sandbox (Months 1-4):** Partner with **MBI Perak** and the **Terra Mineral Lab (TML)** to ingest local "ground-truth" data into the GraphRAG historian.
- **Phase 2: The Regulatory Integration (Months 5-8):** Secure formal recognition from **JMG (Minerals and Geoscience Dept)** as a pre-validated tool for ESG reporting.
- **Phase 3: The Cross-State Expansion (Months 9-18):** Use the Perak success as a case study to onboard Terengganu, Kelantan, and Kedah under the State Sovereign Cluster discount.

Plan Tier	Strategic Pricing Adjustments	Purpose
Enterprise (Sovereign)	RM 500,000/year (Base)	Centralized oversight for state-level mineral governance.
State Sovereign Cluster Discount	20% Rebate when ≥3 sites are onboarded simultaneously.	Encourages states like Perak or Kedah to standardize Zync as the official "Regulatory Truth."
Incentive Structure	"Sovereign Dashboard" included at no cost.	Provides state authorities with real-time, cross-site heatmaps of ESG compliance and yield forecasts.

7.2 Implementation Plan

Phase	Action	Timeline	Target
Soft Launch	Perak Site Pilots (Gerik/Kenering)	Month 1-6	5 Active Sites
Expansion Phase	Kedah, Terengganu, Pahang	Month 6–12	15 Active Sites
Optimization	Multi-State "Sovereign Brain" Hub	Month 12-18	20+ Sites
TOTAL	RM 12.6M investment	18 Months	Significant Market Share (62%)

7.3 Revenue Projections

<i>Metric</i>	<i>Year 1</i>	<i>Year 2</i>	<i>Year 3</i>
<i>Active Sites</i>	<i>8</i>	<i>15</i>	<i>22</i>
<i>Pilot Licenses (RM 5000)</i>	<i>RM 40,000</i>	<i>RM 25,000</i>	<i>RM 0</i>
<i>Annual Subs (RM 36,000)</i>	<i>RM 144,000</i>	<i>RM 432,000</i>	<i>RM 792,000</i>
<i>Sovereign Enterprise (RM 500k)</i>	<i>RM 500,000 (MBI Perak)</i>	<i>RM 1,500,000 (3 States)</i>	<i>RM 3,500,000 (7 States)</i>
<i>Usage Top-Ups (RM 200/100 runs)</i>	<i>RM 15,000</i>	<i>RM 85,000</i>	<i>RM 240,000</i>
<i>Yield-Share (2% of VAR)</i>	<i>RM 0 (Validation Period)</i>	<i>RM 350,000 (Early Adopters)</i>	<i>RM 4,200,000 (Mature Production)</i>
<i>Total Revenue</i>	<i>RM 699,000</i>	<i>RM 2,392,000</i>	<i>RM 8,732,000</i>
<i>Gross Margin %</i>	<i>~82%</i>	<i>~91%</i>	<i>~96%</i>

7.4 Key Financial Milestones

- **Month 4:** Reach **RM 500,000** revenue through the first Sovereign Enterprise contract (MBI Perak).
- **Month 18: Break-Even Point.** Shift from cash-burn to positive contribution margin as with Yield Share (2% of VAR).
- **Year 3:** Reach **Cashflow Positive (+RM 4.5M - 6M)** after overhead, allowing for expansion into downstream magnet design agents.